



LLMs and Social Simulation

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Zeyu Lyu

Computational Humanities and
Social Sciences
Tohoku University

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Introduction

Introduction

Large Language Models (LLMs) are substantially affecting social science research [2, 3].

- Evaluate LLMs from a perspective of Social Science.
- Align LLMs from a perspective of Social Science.
- **Employ LLMs to facilitate research, address issues, and enhance tools in Social Science.**

Agent-based Models (ABMs)

- ABMs allow researchers to explore hypothetical scenarios and identify individual-level patterns that can create macrolevel patterns
- The agents in such a model can be assigned behaviors such as maximizing their own self-interest, and these parameters can be systematically varied in order to identify the range of possible outcomes within the broader social setting.
- Recent work employs more sophisticated agents that can have many characteristics—and follow multiple rules using humanlike decision-making processes.

Agent-based Models (ABMs)

- Conventional ABMs often rely on abstract theories, oversimplified assumptions, and predefined rules.

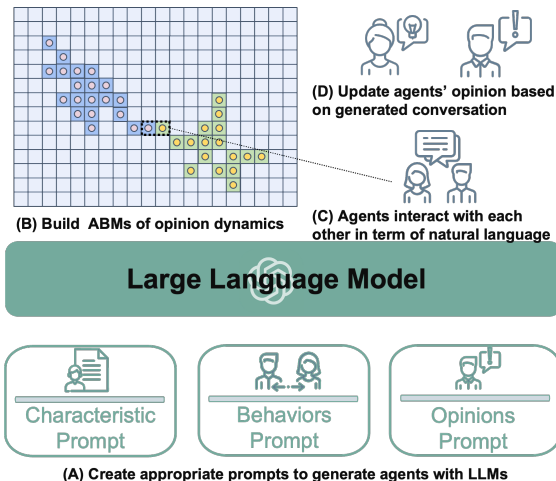
Limitations in Interpretability and Generalization

- **Model Complexity:** Simple agent architecture is not enough to cope with complex tasks.
- **Agent Heterogeneity and Model Adaptability:** Difficult to develop a general agent that can support simulations across environments.
- **Model Interpretability:** Cannot support integrative simulation in real-world problems.

Can LLMs Improve Simulation-Based Research?

- LLMs could help make these simulations more realistic and comprehensive, allowing researchers to model nuanced social interactions and complex decision-making processes.
- This approach could be employed to study topics that are very difficult to examine in real life (such as violent extremism on social media) or to study populations that are very difficult to access.

LLMs-based ABM



Related Works

Generative Agents

- Agents lead their own lives, going through daily routines such as waking up, preparing breakfast, heading to work, and so on.
- They engage with other agents, retain memories, and strategize for the upcoming day.

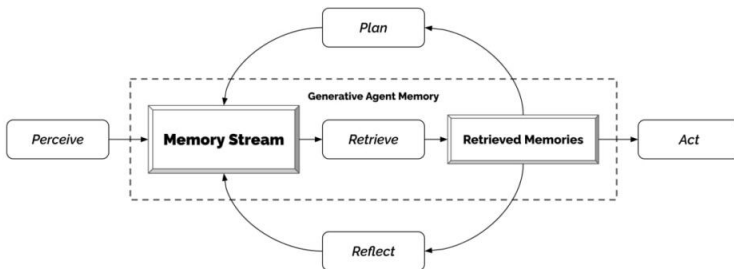


Generative Agents

- The agents are hosted in a sandbox environment modeled after a small town.
- This town comprises houses, a college, shops, parks, cafes, and more. Inside the houses, individual rooms are equipped with items like tables and wardrobes.



Generative Agents



Memory Stream

A special retrieval function takes an agent's current situation as input and returns a subset of events from its memory to pass to the LLM. The score of each memory item depends on three factors:

- Recency: This is an exponentially decaying freshness score, calculated based on the last time an item was accessed.
- Importance: This score comes from the LLM, rated on a scale of 1 to 10. Here, 1 indicates something mundane and minor (like brushing teeth), while 10 represents something significant (like getting a divorce or entering college).
- Relevance: It's determined using cosine similarity between the embeddings (obtained from the LLM) of the query and the memory item.

Reflection - A Higher-level Memory

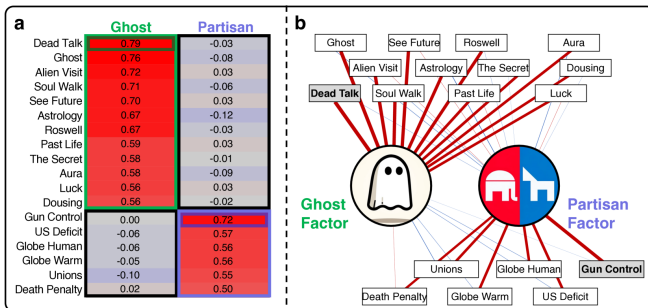
Reflection serves as a secondary, more abstract type of memory generated by the agent.

- To create a reflection via the LLM, the 100 most recent memories of the agent are taken and sent to the LLM with the prompt: *"Given only the information above, what are the 3 most salient high-level questions we can answer about the subjects in the statements?"*.
- Based on the retrieved questions, relevant memories are recalled, and insights are generated using the prompt: *"What 5 high-level insights can you infer from the above statements? (example format: insight (because of 1, 5, 3))"*.

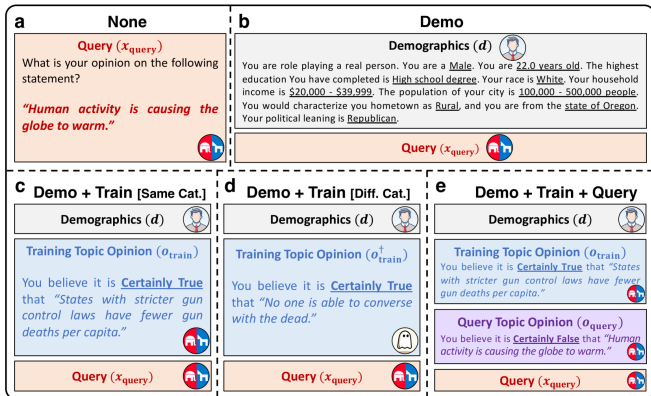
Role-playing and LLMs

- Creating human-like LLM agents is crucial for faithful social simulation.
 - Existing method: Use demographic information to guide role-playing of LLMs.
 - Proposed method: Integrate information from empirically-derived human belief networks.
- Role-playing based on demographic information alone did not align LLM and human opinions, but seeding the agent with a single belief greatly improved alignment for topics related in the belief network, and not for topics outside the network.

Role-playing and LLMs



Role-playing and LLMs



LLMs and Network Generation

- LLMs are useful in generating social network with the proposed prompting methods.
 - “Local” methods: LLM constructs relations for one persona at a time
 - “Global” methods: Construct the entire network at once
- Generated networks based on LLMs match real networks on many characteristics, while might overestimate political homophily.

LLMs and Network Generation

- Real Network: Network describes friendship ties among people with different demographic characteristics and personas.
- Generated Network: Sample $N = 50$ personas and we use the same set of personas for all LLM experiments.

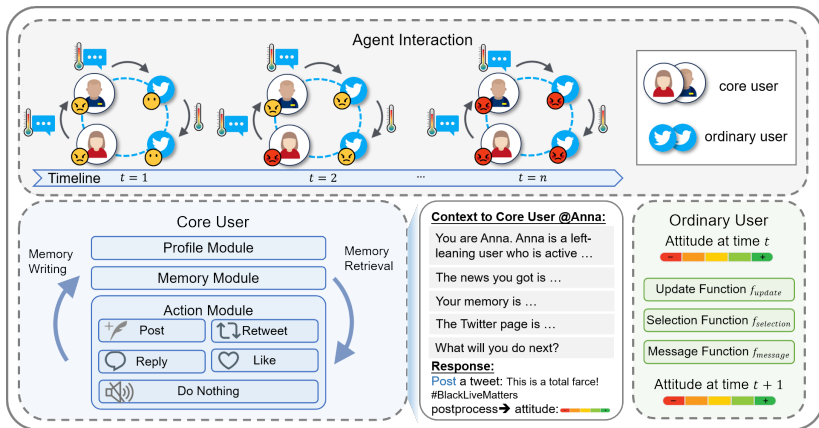
LLMs and Network Generation

Measure	Local	Sequential	Source	Description	Value
Same-group ratio (Eq. 1)	1.851 (0.016)	1.685 (0.024)	Halberstam and Knight (2016)	Twitter	1.404
Isolation index (Eq. 6)	0.720 (0.020)	0.530 (0.027)	Halberstam and Knight (2016)	Twitter	0.403
			Gentzkow and Shapiro (2011)	Voluntary associations	0.145
				Work	0.168
				Neighborhood	0.187
				Family	0.243
Polarization (Eq. 7)	0.639 (0.037)	0.515 (0.041)	Garimella and Weber (2017)	People you trust	0.303
				Political discussants	0.394
				Twitter, follow	0.33-0.42
				Twitter, retweet	0.37-0.41

LLMs and Social Movement Simulation

- Construct a Twitter-like environment and use LLMs-driven agents to replicate individuals' response dynamics following trigger events.
 - Core users are driven by LLMs, while numerous ordinary users are modeled by deductive agent-based models.

LLMs and Social Movement Simulation



Benchmark for Social Movement Simulation

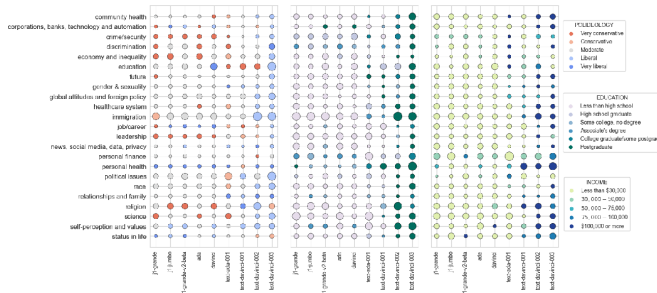
- Collect Twitter datasets related to specific social movements, including users' social networks and tweets during the event period.
- Tweets are annotated with the attitude scores using GPT3.5 and Textblob.
- Select core users by identifying the top 100 most influential individuals based on the number of received retweets.

Dataset	Event	#Users	#Tweets	Time Span
Metoo	E1	1,000	18,638	Oct 15 - Oct 22, 2017
	E2	1,000	13,291	Jan 06 - Jan 13, 2018
Roe	E1	1,000	61,687	May 02 - May 09, 2022
	E2	1,000	59,829	Jun 24 - Jul 01, 2022
BLM	P1	1,000	10,710	May 25 - Jun 01, 2020
	P2	1,000	21,480	Jun 02 - Jun 09, 2020

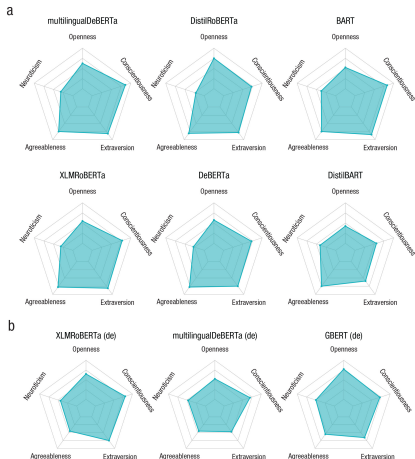
Issues on LLM-based Simulation

LLMs Exhibits Biases

- LLM's responses are more liberal than the general population and reflect those who are younger and have more education.



LLMs Exhibits Biases



Pellert, M. et al. (2024). AI Psychometrics: Assessing the Psychological Profiles of Large Language Models Through Psychometric Inventories? Perspectives on Psychological Science. 26
<https://journals.sagepub.com/doi/10.1177/17456916231214460>.

LLMs Cannot Exhibit Biases

- A key question for social scientists is whether the bias of LLMS is a “bug” or a “feature” for research purposes [1].
- LLMs are usually designed to avoid generating prejudiced, dangerous, or illegal content.
- Biases might be crucial for special research purposes in social science.
 - Use LLMs to identify discrimination and prejudiced opinions.
 - Use LLMs to impersonate biased groups.

How to handle biases in LLMs?

Previous studies typically use prompt engineering

Carefully crafted prompts can guide LLMs to adopt specific roles or personas.

- Prompts can instruct LLMs to embody individuals with particular demographic and psychological traits.
- Prompt can guide LLMs to utilize memory and tools to imitate human behaviors in specific contexts.

How to handle biases in LLMs?

(However) Prompt engineering typically lacks stability and explainability

- A prompt that works well in one instance may not produce the same quality of output in another, even with the same model and similar inputs.
- The relationship between the prompt and the model's output is complex and not always intuitive.

How to handle biases in LLMs?

Proposed Solution: Use fine-tuned LLM to achieve specific role-playing

- A Fine-tuned model is more likely to maintain a consistent persona or role
- Through fine-tuning, even small-scale LLMs can match or outperform large-scale LLMs on specific tasks.

How to employ LLMs in simulation

LLMs API

- The latest LLM models are typically expensive.
- For example, with 500 agents and 20 iterations, each iteration involves an average input of 1,000 tokens and an output of 1,000 tokens.
 - gpt-4o-mini \$ 9.75
 - gpt-4-turbo \$520
 - gemini-1.5-pro \$182

How to employ LLMs in simulation

Local LLM

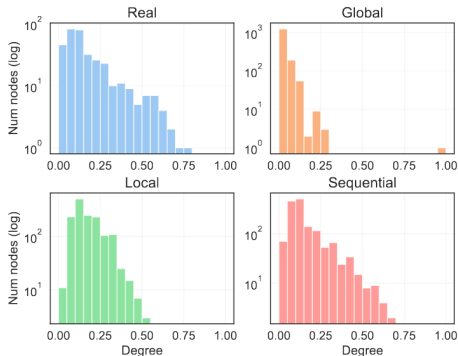
- Trade-off between scalability and speed
 - The RTX A6000 can handle 70B LLMs but is not suitable for 405B LLMs.
- Implementation of local LLMs
 - llama.cpp
 - ollama

How to employ LLMs in simulation

- Design the simulation model consists of LLM-driven agents and rule-based agents.
 - Corresponding theoretical support is needed to justify the use of this design in simulations related to social order.
 - Suitable contexts to apply such design in simulations related to social order.

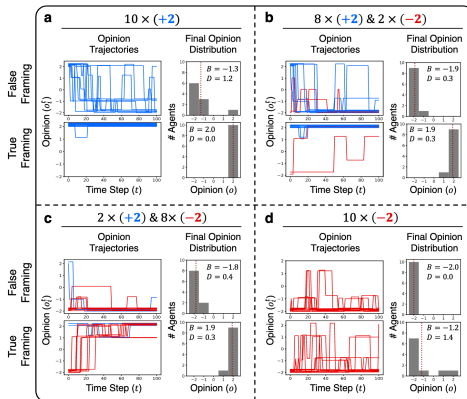
Evaluation metrics

- Network structure metrics.



Evaluation metrics

- Text output of LLM-based agents



How to evaluate LLMs-based simulation

- **Theory-based:** Examine whether the simulation can accurately reproduce phenomena and processes described in theories
- **Empirical data based:** Calibrate simulation results with empirical data